

湖 南 省

高等学校招生数学试题

答案和评分标准

一 说 明

各试题的解答方法和步骤是多种多样的,考生只要解答正确,可评满分;部分解答合理,酌情给分。我们的解答和评分标准,各地可参照研究实行。

二 试题解答和评分标准

一、(理科每小题5分,6小题共30分;文科每小题6分,6小题共36分)

$$1. \text{解法一} \because x^{-\frac{3}{2}} = \frac{27}{8}, \quad 2 \text{ 分}$$

$$\therefore x = \left(\frac{27}{8}\right)^{-\frac{2}{3}} \quad 3 \text{ 分}$$

$$= \left(\frac{8}{27}\right)^{\frac{2}{3}} \quad \text{理4文5}$$

$$= \left(\frac{2}{3}\right)^2 = \frac{4}{9}.$$

理5文6

$$\text{解法二} \because x^{-\frac{3}{2}} = \frac{27}{8},$$

2分

$$\therefore x^{\frac{3}{2}} = \frac{8}{27},$$

理3文4

$$x^{\frac{1}{2}} = \frac{2}{3},$$

理4文5

$$x = \frac{4}{9}.$$

理5文6

注：由 $x^{-\frac{3}{2}} = \frac{27}{8}$ 直接写出 $x = \frac{4}{9}$ 者，给满分。但直接写出

错误答案者，后面部分不给分。

$$2. \text{原式} = 2\lg 6 + 2\sqrt{(\lg 6 - 1)^2}$$

2分

$$= 2\lg 6 + 2(1 - \lg 6) = 2.$$

理5文6

$$3. x \text{ 所取的值应使 } 3 - x \geq 0, \quad x - 2 > 0.$$

3分

$$\therefore 2 < x \leq 3.$$

理5文6

$$4. \text{可化为 } (x+4)(x-7) < 0,$$

2分

$$\therefore -4 < x < 7.$$

理5文5

$$5. \text{证法一 } \cos 15^\circ + \sin 15^\circ = \sin 75^\circ + \sin 15^\circ$$

1分

$$= 2\sin 45^\circ \cos 30^\circ$$

理4文5

$$= 2 \times \frac{\sqrt{2}}{2} \times \frac{\sqrt{3}}{2}$$

$$= \frac{\sqrt{6}}{2}.$$

理5文6

$$\text{证法二 } \cos 15^\circ + \sin 15^\circ$$

$$= \cos 45^\circ \cos 30^\circ + \sin 45^\circ \sin 30^\circ + \sin 45^\circ \cos 30^\circ - \cos 45^\circ \sin 30^\circ \quad 3 \text{ 分}$$

$$= \frac{\sqrt{2}}{2} \times \frac{\sqrt{3}}{2} + \frac{\sqrt{2}}{2} \times \frac{1}{2} + \frac{\sqrt{2}}{2} \times \frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2} \times \frac{1}{2} = \frac{\sqrt{6}}{2}. \quad \text{理5文6}$$

证法三 $\cos 15^\circ + \sin 15^\circ = \sqrt{\frac{2+\sqrt{3}}{4}} + \sqrt{\frac{2-\sqrt{3}}{4}} \quad 2 \text{ 分}$

$$= \frac{1}{2} \sqrt{2+\sqrt{3}} + \frac{1}{2} \sqrt{2-\sqrt{3}} \quad 3 \text{ 分}$$

$$= \frac{1}{2\sqrt{2}} \left(\sqrt{4+2\sqrt{3}} + \sqrt{4-2\sqrt{3}} \right)$$

理4文5

$$= \frac{1}{2\sqrt{2}} \left(\sqrt{3} + 1 + \sqrt{3} - 1 \right) =$$

$$= \frac{\sqrt{6}}{2}. \quad \text{理5文6}$$

6. 设椭圆的方程是 $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, 1 分

则 $\frac{5^2}{a^2} + \frac{0^2}{b^2} = 1$, $\frac{4^2}{a^2} + \frac{\left(\frac{12}{5}\right)^2}{b^2} = 1$. 理3文4

解之, 得 $a=5$, $b=4$. 理4文5

故所求的椭圆方程是 $\frac{x^2}{25} + \frac{y^2}{16} = 1$. 理5文6

二、(理科10分;文12分。)设安排加工甲种部件的 x 人,乙种部件 y 人,丙种部件 z 人,能使加工的三种部件按要求配套。则有

$$\begin{cases} x+y+z=86, & \text{①} & \text{1分} \\ 15x=9z \times 3, & \text{②} & \text{3分} \\ 12y=9z \times 2. & \text{③} & \text{理4文5} \end{cases}$$

由得②, $x = \frac{9}{5}z$, ④ 由③得 $y = \frac{3}{2}z$, ⑤ 理5文6

代入①, 得 $\frac{9}{5}z + \frac{3}{2}z + z = 86$, $z = 86 \times \frac{10}{43} = 20$. 理7文9

代入④, 得 $x = \frac{9}{5} \times 20 = 36$,

代入⑤, 得 $y = \frac{3}{2} \times 20 = 30$. 理9文11

答: 应安排加工甲种部件的36人, 乙种部件的30人, 丙种部件的20人, 才能使三种部件按要求配套。理10文12

三、(理科10分;文科12分。)

解法一(1)若 $x_1 = 0$, 则 $a+1=0$, $\therefore a=-1$. 理3文4

(2) $\therefore x_1 + x_2 = \frac{2a+1}{a} = 0$, $\therefore 2a+1=0$, $a = -\frac{1}{2}$

理6文8

(3) $\therefore x_1 x_2 = \frac{a+1}{a} = 2$, $\therefore a+1=2a$, $a=1$. 理10文12

解法二 $\therefore x = \frac{2a+1 \pm 1}{2a}$, 理3文4

$\therefore x_1 = \frac{a+1}{a}$, $x_2 = 1$. 理4文5

(1) 若 $x_1 = 0$, 则 $\frac{a+1}{a} = 0$, $a = -1$. 理6文7

(2) 若 $x_1 + x_2 = 0$, 即 $x_1 + 1 = 0$, $x_1 = \frac{a+1}{a} = -1$,

$a = -\frac{1}{2}$. 理8文9

(3) 若 $x_1 x_2 = 2$, 则 $x_1 = \frac{a+1}{a} = 2$, $a+1 = 2a$, $a = 1$.

理10文12

四、(理科10分; 文科13分。) 设三个小锡球的半径依次是 r_1, r_2 厘米和 r_3 厘米, 大锡球的半径是 R 厘米, 则有

$4\pi r_1^2 = 36\pi$, $r_1 = 3$; 理2文3

$4\pi r_2^2 = 64\pi$, $r_2 = 4$; 理3文4

$4\pi r_3^2 = 100\pi$, $r_3 = 5$. 理4文5

$\frac{4}{3}\pi r_1^3 + \frac{4}{3}\pi r_2^3 + \frac{4}{3}\pi r_3^3 = \frac{4}{3}\pi R^3$, 理6文8

$R^3 = r_1^3 + r_2^3 + r_3^3 = 3^3 + 4^3 + 5^3 = 216$, 理8文11

$\therefore R = 6$. 理10文13

五、(理科10分; 文科12分。) 设 $\angle A = x - 3d$,

$\angle B = x - d$, $\angle C = x + d$, $\angle D = x + 3d$, 2分

则有 $(x - 3d) + (x - d) + (x + d) + (x + 3d) = 360^\circ$, ①

理5文6

$x + d = 3(x - 3d)$. ② 理6文7

由①: $x = 90^\circ$, 代入②: $d = 18^\circ$. 理8文10

$\therefore \angle A = 90^\circ - 18^\circ \times 3 = 36^\circ$, $\angle B = 90^\circ - 18^\circ = 72^\circ$,

理9文11

$$\angle C = 90^\circ + 18^\circ = 108^\circ, \angle D = 90^\circ + 18^\circ \times 3 = 144^\circ.$$

理10文12

六、(15分) 由正弦定理, 得 $\frac{10}{\sin 45^\circ} = \frac{5\sqrt{6}}{\sin C},$ 3分

$$\therefore \sin C = \frac{5\sqrt{6} \times \frac{1}{\sqrt{2}}}{10} = \frac{\sqrt{3}}{2}, C_1 = 60^\circ, C_2 = 120^\circ$$

6分

故 $A_1 = 75^\circ, A_2 = 15^\circ.$

8分

再由正弦定理, 得 $\frac{a_1}{\sin 75^\circ} = \frac{10}{\sin 45^\circ},$

$$\frac{a_2}{\sin 15^\circ} = \frac{10}{\sin 45^\circ}.$$

10分

可算得 $\sin 75^\circ = \frac{\sqrt{6} + \sqrt{2}}{4}, \sin 15^\circ = \frac{\sqrt{6} - \sqrt{2}}{4}.$

12分

$$\therefore a_1 = 5\sqrt{3} + 5.$$

13分

$$a_2 = 5\sqrt{3} - 5.$$

14分

故本题有两解: $\begin{cases} C_1 = 60^\circ, \\ A_1 = 75^\circ, \\ a_1 = 5\sqrt{3} + 5; \end{cases} \begin{cases} C_2 = 120^\circ, \\ A_2 = 15^\circ, \\ a_2 = 5\sqrt{3} - 5. \end{cases}$ 15分

注: 本题只求得一组解者扣5分.

七、(15分) $\angle A + \angle C = \angle B + \angle D = 180^\circ.$ 1分

又 $\angle A = 60^\circ, \angle B = 90^\circ, \therefore \angle C = 120^\circ,$

$$\angle D = 90^\circ.$$

2分

以下列两种解法:

解法一 延长 BC 、 AD 交于 E , 则 $\angle E = 30^\circ$, $\angle ECD = 60^\circ$,
 $\angle EDC = 90^\circ$, 9分

$$\therefore AE = 4, CE = 2, BE = \sqrt{4^2 - 2^2} = 2\sqrt{3},$$

$$DE = \sqrt{2^2 - 1} = \sqrt{3}. \quad 13分$$

$$\text{故 } BC = BE - CE = 2\sqrt{3} - 2,$$

$$DA = AE - DE = 4 - \sqrt{3}. \quad 15分$$

解法二 设 $BC = x$, $DA = y$, 则有

$$x^2 + 2^2 = y^2 + 1^2, \quad 5分$$

$$x^2 + 1^2 - 2x \cos 120^\circ = y^2 + 2^2 - 4y \cos 60^\circ. \quad 8分$$

$$\text{化简, 得 } x^2 - y^2 + 3 = 0, \textcircled{1} \quad x^2 - y^2 + x + 2y - 3 = 0. \textcircled{2} \quad 9分$$

$$\textcircled{2} - \textcircled{1}, \text{ 并化简, 得 } x = 6 - 2y, \textcircled{3}$$

$$\textcircled{3} \text{ 代入 } \textcircled{1}, \text{ 得 } y^2 - 8y + 13 = 0. \quad 12分$$

$$\therefore y = 4 \pm \sqrt{3}, \text{ 代入 } \textcircled{3}, \text{ 得 } x = -2 \mp 2\sqrt{3}. \quad 13分$$

$$\because x > 0, \therefore \begin{cases} x = 2\sqrt{3} - 2, \\ y = 4 - \sqrt{3}. \end{cases} \quad 15分$$

参考题 (不记分, 但要作出评语)

一、用数学归纳法或其他方法证明

$$1^4 + 2^4 + 3^4 + \dots + n^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30}.$$

证明(1) $\because n = 1$ 时, 等式的左边 $= 1^4 = 1$,

$$\text{等式的右边} = \frac{1 \cdot 2 \cdot 3 \cdot 5}{30} = 1,$$

$\therefore n = 1$ 时, 等式成立.

(2) 假设 $n = k$ 时等式成立, 即假设

$$1^4 + 2^4 + 3^4 + \cdots + k^4 = \frac{k(k+1)(2k+1)(3k^2+3k-1)}{30} \quad ①.$$

由此证明 $n = k+1$ 时, 等式也成立, 即要证明

$$\begin{aligned} & 1^4 + 2^4 + 3^4 + \cdots + (k+1)^4 = \\ &= \frac{(k+1)[(k+1)+1][2(k+1)+1][3(k+1)^2+3(k+1)-1]}{30}, \end{aligned}$$

$$\begin{aligned} \text{或 } & 1^4 + 2^4 + 3^4 + \cdots + (k+1)^4 = \\ &= \frac{(k+1)(k+2)(2k+3)(3k^2+9k+5)}{30}. \end{aligned}$$

根据①式, 可得

$$\begin{aligned} & 1^4 + 2^4 + 3^4 + \cdots + (k+1)^4 = \\ &= \frac{k(k+1)(2k+1)(3k^2+3k-1)}{30} + (k+1)^4 \\ &= \frac{k+1}{30} [k(2k+1)(3k^2+3k-1) + 30(k+1)^4] \\ &= \frac{k+1}{30} (6k^4 + 39k^3 + 91k^2 + 89k + 30) \\ &= \frac{(k+1)(k+2)(2k+3)(3k^2+9k+5)}{30}. \end{aligned}$$

这就证明了 $n = k+1$ 时, 等式也成立.

所以, 对于任意自然数 n , 等式都成立.

$$6 + 39 + 91 + 89 + 30 \mid -2$$

$$\begin{array}{r} -12 - 54 - 74 - 30 \\ \hline 6 + 27 + 37 + 15 \mid + 0 \mid -\frac{3}{2} \end{array}$$

$$\frac{-9-27-15}{2\left|\frac{6+18+10}{3+9+5}+0\right|}$$

二、计算下列积分:

$$(1) \int_0^1 x e^{-x} dx; \quad (2) \int_0^a x^2 \sqrt{a^2 - x^2} dx.$$

解(1) 设 $u = x$, $dv = e^{-x} dx$, 则 $v = -e^{-x}$.

$$\begin{aligned} \int_0^1 x e^{-x} dx &= -x e^{-x} \Big|_0^1 + \int_0^1 e^{-x} (x)' dx \\ &= -1 \cdot e^{-1} - e^{-x} \Big|_0^1 = -\frac{1}{e} - e^{-1} + 1 = 1 - \frac{2}{e}. \end{aligned}$$

(2) 设 $x = a \sin \theta$, 则 $dx = a \cos \theta d\theta$,

$$\begin{aligned} \int_0^a x^2 \sqrt{a^2 - x^2} dx &= \int_0^{\frac{\pi}{2}} a^2 \sin^2 \theta \sqrt{a^2 - a^2 \sin^2 \theta} \cdot a \cos \theta d\theta \\ &= \int_0^{\frac{\pi}{2}} a^4 \sin^2 \theta \cos^2 \theta d\theta = \frac{a^4}{4} \int_0^{\frac{\pi}{2}} \sin^2 2\theta d\theta = \frac{a^4}{4} \int_0^{\frac{\pi}{2}} \frac{1 - \cos 4\theta}{2} d\theta \\ &= \frac{a^4}{4} \left[\frac{1}{2} \theta - \frac{1}{8} \sin 4\theta \right]_0^{\frac{\pi}{2}} = \frac{a^4}{4} \left[\frac{\pi}{4} - \frac{1}{8} \sin 2\pi \right] = \frac{\pi a^4}{16}. \end{aligned}$$